

Benchmark results compiled with -O2: mean time (microseconds)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	164.406	155.318	1249.651	4.721	147.058	294.129	264.201	90.061	190.162	1081.089	4.759	4.737	–	–	–
countdown.deep.10000	1284.844	581.568	2316.045	2789.041	147.347	290.34	166.588	171.422	195.212	3556.969	4.794	4.783	66.154	–	–
catch.shallow.10000	274.905	36.443	1137.443	21.092	1090.236	380198.9	248.139	75.486	849.741	851.067	21.151	20.624	–	–	–
catch.deep.10000	3239.011	35.358	6648.266	1719.033	1091.939	396626.6	176.196	239.938	853.506	855.61	21.048	21.017	169.922	–	–
local.shallow.10000	950.815	462.506	2185.586	5.011	554.755	380601.8	205.435	138.153	799.632	796.553	4.971	4.93	–	–	–
local.deep.10000	3856.278	467.383	5130.447	4.998	555.713	393730.3	203.836	223.885	805.193	805.945	4.981	4.972	209.6	–	–
nondet.shallow.32	2008.012	3298.067	7642.203	275.058	–	25147.388	804.602	938.941	1767.65	1769.424	224.004	248.761	–	358.963	363.75
nondet.deep.32	5545.209	5839.631	7719.056	2562.375	–	51372	786.139	634.224	5768.506	5789.041	248.453	246.12	1021.765	64736.7	64256.15

Benchmark results compiled with -O2: mean time (% of fastest)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	3482.7%	3290.2%	26472.1%	100.0%	3115.2%	6230.7%	5596.7%	1907.8%	4028.3%	22901.4%	100.8%	100.3%	–	–	–
countdown.deep.10000	26861.2%	12158.4%	48419.7%	58308.3%	3080.5%	6069.9%	3482.7%	3583.8%	4081.1%	74362.7%	100.2%	100.0%	1383.0%	–	–
catch.shallow.10000	1332.9%	176.7%	5515.0%	102.3%	5286.1%	1843436.7%	1203.1%	366.0%	4120.1%	4126.5%	102.6%	100.0%	–	–	–
catch.deep.10000	15411.3%	168.2%	31632.7%	8179.2%	5195.5%	1887165.6%	838.3%	1141.6%	4061.0%	4071.0%	100.1%	100.0%	808.5%	–	–
local.shallow.10000	19287.9%	9382.2%	44336.0%	101.7%	11253.6%	7720747.2%	4167.4%	2802.5%	16221.0%	16158.6%	100.8%	100.0%	–	–	–
local.deep.10000	77552.3%	9399.4%	103176.7%	100.5%	11175.8%	7918175.0%	4099.3%	4502.5%	16193.0%	16208.1%	100.2%	100.0%	4215.2%	–	–
nondet.shallow.32	896.4%	1472.3%	3411.6%	122.8%	–	11226.3%	359.2%	419.2%	789.1%	100.0%	100.0%	111.1%	–	160.2%	162.4%
nondet.deep.32	2253.1%	2372.7%	3136.3%	1041.1%	–	20872.7%	319.4%	257.7%	2343.8%	2352.1%	100.9%	100.0%	415.1%	26302.9%	26107.6%

Benchmark results compiled with -O2: allocated bytes

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	1760143	1600143	13361522	24	1121566	2087371	2480238	1040119	3441159	14241105	24	24	–	–	–
countdown.deep.10000	17760450	10400193	37363460	27121108	1136511	2088075	2000190	3920270	3442686	41443874	24	24	1200099	–	–
catch.shallow.10000	4069590	65595	7764881	65585	7846336	413419395	1687790	723951	6081377	6081377	65585	65585	–	–	–
catch.deep.10000	16870401	66058	55153625	16165074	7864137	414232477	1527782	3367870	6083937	6083937	65585	65585	1527775	–	–
local.shallow.10000	7205041	4793285	12629896	24	4326241	409594163	1669446	1029423	5513561	5513562	24	24	–	–	–
local.deep.10000	16005986	4793724	30232160	24	4341353	410400203	1589470	2549550	5515808	5515809	64	64	1589470	–	–
nondet.shallow.32	13454089	15889891	43492474	3606902	–	111260594	1531314	1532744	22646044	22646041	1987326	2003982	–	3664086	3664078
nondet.deep.32	67560022	45645998	43569831	26810636	–	383225982	1531394	1535116	68636385	68636396	1969094	2010670	7408464	469818460	469818664

Benchmark results compiled with -O2: allocated bytes (% of least allocated)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	7333929.2%	6667262.5%	55673008.3%	100.0%	4673191.7%	8697379.2%	10334325.0%	4333829.2%	14338162.5%	59337937.5%	100.0%	100.0%	–	–	–
countdown.deep.10000	74001875.0%	43334137.5%	155681083.3%	113004616.7%	4735462.5%	8700312.5%	8334125.0%	16334458.3%	14344525.0%	172682808.3%	100.0%	100.0%	5000412.5%	–	–
catch.shallow.10000	6205.1%	100.0%	11839.4%	100.0%	11963.6%	630356.6%	2573.4%	1103.8%	9272.5%	9272.5%	100.0%	100.0%	–	–	–
catch.deep.10000	25723.0%	100.7%	84094.9%	24647.5%	11990.8%	631596.4%	2329.5%	5135.1%	9276.4%	9276.4%	100.0%	100.0%	2329.5%	–	–
local.shallow.10000	30021004.2%	19972920.8%	52624566.7%	100.0%	18026004.2%	1706642345.8%	6956025.0%	4289262.5%	22973170.8%	22973170.0%	100.0%	100.0%	–	–	–
local.deep.10000	66691608.3%	19973850.0%	125967333.3%	100.0%	18088970.8%	1710000845.8%	6622791.7%	10623125.0%	22982533.3%	22982537.5%	266.7%	266.7%	6622791.7%	–	–
nondet.shallow.32	878.6%	1037.7%	2840.2%	235.5%	–	7265.7%	100.0%	100.1%	1478.9%	1478.9%	129.8%	129.8%	–	239.3%	239.3%
nondet.deep.32	4411.7%	2980.7%	2845.1%	1750.7%	–	25024.6%	100.0%	100.2%	4482.0%	4482.0%	128.6%	128.6%	131.3%	483.8%	30679.1%

Benchmark results compiled with -O0: mean time (microseconds)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	798.538	509.13	1733.018	1749.357	740.727	896.962	1614.533	1625.73	845.102	2174.377	654.79	565.603	–	–	–
countdown.deep.10000	3004.258	978.679	2890.541	10750.331	744.925	885.597	1677.138	3247.911	1447.842	5357.541	5238.784	5922.169	1689.268	–	–
catch.shallow.10000	1083.237	310.476	2808.698	3467.633	2154.685	379758.1	963.012	1090.171	1661.942	1658.47	256.513	226.925	–	–	–
catch.deep.10000	8351.972	317.406	7077.556	16239.5	2193.773	412260	1033.715	2981.111	1670.427	1655.556	3962.116	4114.517	1119.016	–	–
local.shallow.10000	4518.962	674.51	2651.992	1430.328	1072.911	416023	629.987	666.496	1406.982	1399.893	352.422	329.673	–	–	–
local.deep.10000	13442.725	678.417	5617.95	5973.106	1082.439	394305.5	636.916	1640.293	1414.94	1409.345	818.892	1608.341	651.544	–	–
nondet.shallow.32	7761.631	6054.539	13238.856	2138.505	–	28539.737	4167.837	4231.028	3094.095	3069.472	2469.588	1978.713	–	7458.944	7026.009
nondet.deep.32	15788.212	8541.178	22926.463	10690.034	–	54767.725	4287.117	9690.469	7770.347	7847.113	6804.812	7173.219	4540.744	80644.95	80662.85

Benchmark results compiled with -O0: mean time (% of fastest)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	156.8%	100.0%	340.4%	343.6%	145.5%	176.2%	317.1%	319.3%	166.0%	427.1%	128.6%	111.1%	–	–	–
countdown.deep.10000	403.3%	131.4%	388.0%	1443.1%	100.0%	118.9%	225.1%	436.0%	194.4%	719.2%	703.3%	795.0%	226.8%	–	–
catch.shallow.10000	477.4%	136.8%	1237.7%	1528.1%	99.5%	167349.2%	424.4%	480.4%	732.4%	730.8%	113.0%	100.0%	–	–	–
catch.deep.10000	2631.3%	100.0%	2229.8%	5116.3%	691.2%	129884.3%	325.7%	939.2%	526.3%	521.6%	1248.3%	1296.3%	352.6%	–	–
local.shallow.10000	1370.7%	204.6%	804.4%	433.9%	325.4%	126192.8%	191.1%	202.2%	426.8%	424.6%	106.9%	100.0%	–	–	–
local.deep.10000	2110.6%	106.5%	882.1%	937.8%	170.0%	61908.6%	100.0%	257.5%	222.2%	221.3%	128.6%	252.5%	102.3%	–	–
nondet.shallow.32	392.3%	306.0%	669.1%	108.1%	–	1442.3%	210.6%	213.8%	156.4%	155.1%	124.8%	100.0%	–	377.0%	355.1%
nondet.deep.32	368.3%	199.2%	534.8%	249.4%	–	1277.5%	100.0%	226.0%	181.2%	183.0%	158.7%	167.3%	105.9%	1881.1%	1881.5%

Benchmark results compiled with -O0: allocated bytes

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	10640929	6080468	17281855	11920929	5202068	6268896	10801354	11601186	7041755	19361586	4080228	2880382	–	–	–
countdown.deep.10000	38643708	14881329	41284522	63365159	5218153	6269549	11213138	15371572	9444800	48965273	30882771	30882818	11365118	–	–
catch.shallow.10000	7913899	2571780	22351497	13075266	12013482	415979980	5433985	6393802	8161793	8161738	1673342	1593382	–	–	–
catch.deep.10000	33116289	2572596	57554513	64806117	12032369	416832659	5845705	10164144	8165094	8164585	16926257	18046442	5837705	–	–
local.shallow.10000	14981585	5771833	14008651	7688311	6726752	411559139	4008436	4648292	7193841	7193844	2327804	2327860	–	–	–
local.deep.10000	34184834	5772650	31611412	28490620	6742819	412321159	4020188	6818551	7197129	7196625	5048097	9048609	4012065	–	–
nondet.shallow.32	53827095	34531877	81643735	18238378	–	131920524	24814452	29682938	30881955	30881955	17396960	12390832	–	74123042	69116545
nondet.deep.32	137695415	64288357	200681246	63404958	–	403887019	26178251	42408117	80928928	80928649	43632105	39124526	28604449	543614271	539102643

Benchmark results compiled with -O0: allocated bytes (% of least allocated)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	369.4%	211.1%	600.0%	413.9%	180.6%	217.6%	375.0%	402.8%	244.5%	672.2%	141.7%	100.0%	–	–	–
countdown.deep.10000	740.6%	285.2%	791.2%	1214.3%	100.0%	120.1%	214.9%	294.6%	181.0%	938.4%	591.8%	591.8%	217.8%	–	–
catch.shallow.10000	496.7%	161.4%	1402.8%	820.6%	754.0%	26106.7%	341.0%	401.3%	512.2%	512.2%	105.0%	100.0%	–	–	–
catch.deep.10000	1287.3%	100.0%	2237.2%	2519.1%	467.7%	16202.8%	227.2%	395.1%	317.4%	317.4%	657.9%	701.5%	226.9%	–	–
local.shallow.10000	643.6%	248.0%	601.8%	330.3%	289.0%	17680.1%	172.2%	199.7%	309.0%	309.0%	100.0%	100.0%	–	–	–
local.deep.10000	852.1%	143.9%	787.9%	710.1%	168.1%	10277.0%	100.2%	170.0%	179.4%	179.4%	125.8%	225.5%	100.0%	–	–
nondet.shallow.32	434.4%	278.7%	658.9%	147.2%	–	1064.7%	200.3%	239.6%	249.2%	249.2%	140.4%	100.0%	–	598.2%	557.8%
nondet.deep.32	526.0%	245.6%	766.6%	242.2%	–	1542.8%	100.0%	162.0%	309.1%	309.1%	166.7%	149.5%	109.3%	2076.6%	2059.4%